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Anaerobic digestion of waste biomass from the production of L-cystine in suspended-growth bioreactors

Juan José Chávez-Fuentes, Miroslav Hutňan, Igor Bodík, Ronald Zakhar and Marianna Czölderová

ABSTRACT

Waste biomass from the industrial production of the amino acid L-cystine contains above-average concentrations of organic pollutants and significant concentrations of nitrogen and sulfur. The specific biogas production (SBP) of waste biomass was monitored in parallel suspended-growth laboratory anaerobic bioreactors. After severe inhibition was observed, three different procedures were applied to inhibited reactor sludge to counter-attack the inhibitory effects of sulfides, respectively hydrogen sulfide: micro-aeration, dilution with water and precipitation by ferrous iron cations. The performance of bioreactors was weekly monitored. Organic loading rates (as chemical oxygen demand, COD) ranged from 1.07 to 1.97 g L^{-1} d^{-1}. At the end of the experimentation, SBP averaged 217, 300 and 320 l kg^{-1} COD with a methane content of 21%, 52% and 54%; specific sludge production averaged 133, 111 and 400 g total solids kg^{-1} COD, and inhibition was 49%, 27% and 25%; for the applied procedures of micro-aeration, dilution and precipitation respectively.

Key words | amino acids, biogas, cystine, inhibition, sulfate, sulfide

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INTRODUCTION

The present work was oriented to analyze the energetic potential of waste biomass from the industrial production of L-cystine, which has significant concentrations of organic pollutants, sulfates and nitrogen through its anaerobic digestion in suspended-growth anaerobic bioreactors, incorporating a procedure to counter-attack the inhibitory effects of sulfides on the anaerobic digestion of waste biomass.

Problems linked to anaerobic digestion of organic materials with high content of sulfates

When waste materials with high content of sulfates are loaded in an anaerobic bioreactor, the organic matter is reduced by both sulfate- and methane-reducing bacteria. In the presence of sulfates, acidogenic, acetogenic and methanogenic Archaea compete with sulfate-reducing bacteria (SRB) for organic substrate. In general, SRB are thermodynamically and kinetically stronger competitors than methanogens. The result of such kinetic rivalry among micro-organisms is important, as it determines the production rate of sulfide and methane (Chio & Rim 1991; Mizuno *et al.* 1998; Hulshoff Pol *et al.* 2001).

SRB benefit from anaerobic respiration due to the fact that sulfates act as electron acceptors providing more energy for microbial growth than carbon dioxide. SRB have more affinity to hydrogen and acetate, and have a higher growth rate than methanogenic Archaea. The affinity of SRB is higher with lower chemical oxygen demand (COD) to SO_4^{2-} ratios. In a ratio greater than 0.67, there is theoretically enough organic substrate (COD) for the entire reduction of sulfates (Omil *et al.* 1998). For industrial purposes, COD/SO_4^{2-} ratios should be greater than 10 to guarantee a stable biogas production (Rinzema & Lettinga 1988; O'Flaherty *et al.* 1998). Summarizing, it is evident that the main problem linked to the anaerobic treatment of organic materials with high content of sulfates lies in the fact that the side-growth of SRB and the subsequent production of reduced forms of sulfur compounds such as sulfides will inhibit the process and lower the production rate of methane (Holmer & Kristensen 1994). Some procedures can be applied to promote the removal of sulfides in reactor sludge. A well-known procedure to prevent the formation and further inhibition by sulfides and free ammonia is the dilution of the waste

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stream, although in general this approach is considered undesirable from a practical point of view, because it increases the total volume of waste stream that must be treated. Other removal methods include physico-chemical procedures such as stripping; chemical reactions such as coagulation, oxidation or precipitation; or biological conversions such as partial oxidation of sulfides into elemental sulfur (Bekmezci *et al.* 2011). Furthermore, the adaptation of the methanogenic Archaea to non-dissociated hydrogen sulfide and free ammonia, particularly in bioreactors with fixed biomass, could increase its tolerance (Chen *et al.* 2008).

Waste biomass from the industrial production of L-cystine

The amino acid L-cystine (CAS: 56-89-3, E921) is synthesized by the company Biotika a.s. in central Slovakia through a fermentation process that roughly consists of three steps: cultivation of the specific productive microbial culture, addition of a thiosulfate-rich solution and nutrients into the medium, and the production phase of L-cystine. When the desired conversion rate is achieved, the downstream process takes place. Thus, the fermentation product is sterilized and L-cystine is crystallized and separated, conditioned and dried. The final product is a fine crystallized amino acid, ready for its packing and further delivery. The production of L-cystine generated a revenue from sales €11.8 million in 2013 (Biotika 2013).

During downstream processes, however, two main by-products are formed: waste biomass and mother liquors. Waste biomass is not suitable for conventional aerobic treatment systems as it would represent higher aeration costs; conversely, it has an interesting energy potential and significant fertilizing properties. Nevertheless, due to its pungent odor and conditioning costs, it is not always possible to use it directly as a fertilizer. An effective alternative for the treatment of waste biomass from the production of cystine can be through anaerobic digestion with suppression of the inhibitory effects of non-dissociated hydrogen sulfide and free ammonia.

METHODS AND TECHNIQUES

Feedstock

Waste biomass from the industrial production of L-cystine is a yellow fluid suspension containing above-average concentrations of organic pollutants and significant concentrations of sulfates and nitrogen (Figure 1 (left)). In 2013, about 17,000 ton of this organic material was produced within the company during downstream processes. Selected parameters of waste biomass can be seen in Table 1 for non-filtrated and filtrated samples, respectively.

For this study, three samples provided by Biotika, a.s. were analyzed. The characteristics of samples 2 and 3 were quite similar. However, the concentrations of sulfates and nitrogen in sample 1 were much higher than in samples 2 and 3. These differences were caused by operational



Figure 1 | Left: waste biomass from the production of L-cystine; right: parallel bioreactors.

Table 1 | Selected parameters of substrate

Non-filtrated samples	TS g L^{-1}	VS %	COD mg L^{-1}	NH_4-N mg L^{-1}	N_{Tot} mg L^{-1}	PO_4-P mg L^{-1}	pH	ORP mV
Sample 1	107	51	98,000	1,693	14,400	210	4.12	−200
Sample 2	97	72	150,000	2,095	11,000	300	3.96	−162.7
Sample 3	110	80	136,000	1,748	8,180	314	4.28	−259
Filtrated samples	**DIS g L^{-1}**	**COD mg L^{-1}**	**NH_4-N mg L^{-1}**	**N_{Tot} mg L^{-1}**	**PO_4-P mg L^{-1}**	**SO_4^{2-} mg L^{-1}**	**COD/SO_4^{2-} –**	
Sample 1	46	41,000	1,058	8,290	170	26,700	1.5	
Sample 2	25	92,600	1,070	5,410	230	4,510	20.5	
Sample 3	22	86,500	1,391	3,250	235	3,070	28.2	

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changes in the production line of the company, which can certainly happen in some periods of the year. The samples were taken from the collecting tank situated next to the separation stage of the product (raw L-cystine) from waste biomass. The volume of samples increased from 1 (for sample 1) to 5 liters (samples 2 and 3), due to the fact that there were no substantial changes when the samples were stored at a temperature of 4 °C.

Through elemental analysis using a Vario Macro Cube (Elementar Analysensysteme), the elemental composition of waste biomass (samples 2 and 3) was determined as follows: C 4.41%, N 0.72%, S 0.41% with ratios C:N = 6.1 and C:S = 10.8 for sample 2, and C 4.8%, N 0.94%, S 0.44% with ratios C:N = 5.1 and C:S = 10.9 for sample 3.

Experimental set-up and procedures

At first, the inhibitory effect of sulfides was studied through a biochemical methane test (BMP). For this purpose, waste biomass (sample 1) was used to test the inhibitory effects that sulfides can have on the anaerobic process in three parallel laboratory continuous-stirred bioreactors under mesophilic conditions. As inoculum for anaerobic reactors, anaerobically stabilized excess sludge was used (total solids (TS) = 17 g l^{-1}, volatile solids (VS) = 10.2 g l^{-1}, COD = 209 mg l^{-1}, NH_4-N = 400.6 mg l^{-1}) from the wastewater treatment plant Devínska Nová Ves, in Bratislava. After severe inhibition was observed, three different procedures were implemented for each reactor, in order to counter-attack the inhibitory effect of sulfides in reactor sludge: micro-aeration, dilution with tap water and precipitation by ferrous iron cations. During the long-term operation of bioreactors, waste biomass samples 2 and 3 were dosed and sludge parameters were weekly monitored in order to characterize the performance of bioreactors.

In Figure 1 (right), the bioreactors used in this study are shown. The effective volume of bioreactors was 1.4 l. They were continually mixed (100 rpm) and operated at a temperature of 37 ± 0.5 °C.

Analytical methods

During the experimentation, pH and redox potential (ORP) were measured with a portable multi-probe HQ40d (Hach) in reactor sludge. COD, NH_4-N, PO_4-P and SO_4^{2-} concentrations were determined according to ISO international standards. Volatile fatty acids (VFA) were measured by titration with a 0.05 M solution of H_2SO_4. TS, VS, and dissolved inorganic salts (DIS) were determined gravimetrically according to *Standard Methods for the Examination of Water and Wastewater* (APHA/AWWA/WEF 1998). The production of biogas was monitored with drum-type gas flow meters, model TCM 143/10. The composition of biogas was measured with an infrared gas analyzer GA2000 (Geotechnical Instruments). The analyzer monitors the concentration of CH_4, CO_2, O_2 and N_2. It is also capable of measuring the concentration of H_2S within the range of 0–5,000 ppm_v.

RESULTS AND DISCUSSION

Based on the COD/SO_4^{2-} ratio of waste biomass (Table 1), it can be assumed that waste biomass has enough organic matter to achieve a complete reduction of sulfates to sulfides and still produce a valuable amount of biogas. The first decrease in biogas production observed during the inhibition test was due to the consumption of COD by SRB for the reduction of sulfates and for their own growth. Later, severe inhibition was caused by the accumulation of sulfides, respectively hydrogen sulfide in reactor sludge. During the long-term anaerobic digestion of waste biomass, different inhibition rates caused by hydrogen sulfide and free ammonia were observed.

Inhibitory effect of sulfides on the anaerobic digestion of waste biomass

The BMP test was realized in three parallel continuous-stirred tank reactors with repeatable organic loads of 3.5 g COD L^{-1} (biomass waste sample 1) and 0.35 g $NaHCO_3$ L^{-1} as pH stabilizer. Dosage of reactors was executed four times, after daily biogas production rate declined to less than 1% of the total biogas production up until that time (Verein Deutscher Ingenieure 2006). This inhibition test showed a remarkable drop in specific biogas yields after every dosage (1st dosage – 37.5 mL/mL, 2nd dosage – 20.44 mL/mL, 3rd dosage – 16.48 mL/mL and 4th dosage – 8.8 mL/mL) and an increment in the concentrations of hydrogen sulfide, indicating an increasing inhibition of the process by forming sulfides. After 18 days, severe inhibition of the process was observed and the biogas production rate was reduced to zero. From that moment, three different procedures were applied to inhibited reactor sludge to counter-attack the inhibitory effect of sulfides. The results of such procedures are presented below.

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Performance of bioreactors during long-term anaerobic digestion of waste biomass with the addition of a procedure to counter-attack inhibition by sulfides

Procedure 1: micro-aeration of reactor sludge (R1)

The inhibitory effect of sulfides was suppressed by the partial oxidation of sulfides to elementary sulfur (S°). Micro-aeration was carried out in an external bypass for the whole volume of reactor sludge before daily dosage of waste biomass (samples 2 and 3). The ORP parameter was used as an indicator of aeration intensity (allowing a maximum increase of 25 mV) (Khanal & Ju-Chang 2003). The concentration of sulfides in reactor sludge was reduced by 98% (filtrate). The content of hydrogen sulfide in biogas decreased to about 2,000 ppm_V for organic loading rates (OLRs) lower than 1 g COD L^{-1} d^{-1}.

Micro-aeration also strengthens the stripping of free ammonia. After the first 5 weeks of operation, free ammonia concentration decreased to values lower than 100 mg L^{-1} and only increased for short times when OLR was raised. NH_4-N concentrations moved around 2,000 mg L^{-1} in sludge. The specific biogas production (SBP) during the operation of this reactor was the most unstable of the studied parameters, ranging from 80 to 400 L kg^{-1} COD with a methane content ranging from 9 to 34%. Toward the end of the experimentation, SBP rates averaged 217 L kg^{-1} COD with a methane-content of 21%. The specific production of sludge (SPS) was 133 g TS kg^{-1} COD. Average sludge parameters are shown in Table 1. Monitored inhibition was 49% in respect to measured values during the BMP test.

Micro-aeration comes out as an interesting option to counter-attack sulfide and free ammonia inhibition, but it must be optimized. Two main complications were observed during the operation of the reactor with micro-aeration: firstly, the adsorption of sulfur onto the inner surface of the reactor, pipelines and other reactor equipment and its subsequent accumulation which long-term could lead to many operational problems; secondly, the dilution of biogas occasioned by molecular nitrogen in air. Measured biogas composition indicated that high amounts of nitrogen got into biogas during aeration of reactor sludge. Molecular nitrogen reduces methane content in biogas and subsequently lowers the biogas's heat of combustion. Moreover, aeration of reactor sludge requires special safety regulations as some CH_4/O_2 ratios can create flammable mixtures (Figure 2).

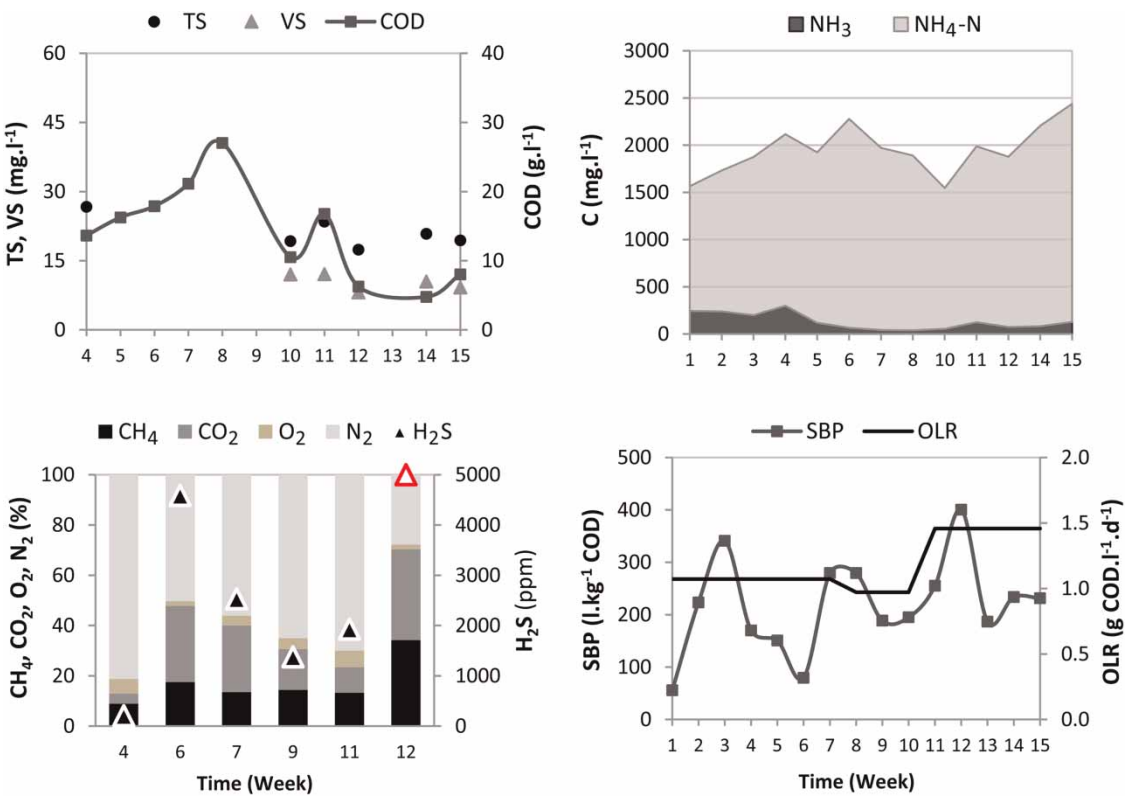


Figure 2 | Micro-aeration of reactor sludge (R1). Variation of selected parameters in bioreactor.

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Procedure 2: dilution of substrate with tap water (R2)

Daily dosage of waste biomass was diluted with water at a 1:1 ratio, during the entire operation of the bioreactor.

Through this procedure, the reactor's retention time was modified and thus, also, parameters such as concentration of micro-organisms, nutrients and inhibitors in reactor sludge. NH_4-N concentration in filtrate was maintained at acceptable operational values (NH_4-N averaged 1,350 mg L^{-1}), but overpassed 1,500 mg l^{-1} when OLR was raised to 1.46 g COD L^{-1} d^{-1}. Free ammonia remained at low values averaging a concentration of 60 mg L^{-1} during the entire operation of the bioreactor. COD concentration of waste biomass was reduced by 99% (filtrate). The results indicate a more stable production of biogas than that of micro-aeration, averaging an SBP of 260 L kg^{-1} COD d^{-1}, with 55% methane. Moreover, process parameters such as TS, VS, VFA, DIS, PO_4-P and pH were stable.

Owing to the fact that dilution of waste biomass is not directly related to chemical changes of sulfurous compounds, the concentration of hydrogen sulfide was much higher in both reactor sludge and biogas than for the other two studied procedures. Analyzed biogas showed concentrations of hydrogen sulfide much higher than 5,000 ppm_v (estimated at 18,000 ppm_v) during the entire operation of the reactor. Therefore, this procedure would require a subsequent H_2S-removal process prior to its combustion in a cogeneration unit. High concentrations of hydrogen sulfide may inhibit the anaerobic process at long-term. Another disadvantage of the process is the amount of anaerobic sludge at the outlet of the reactor, which may double in quantity, as dilution was performed at a 1:1 ratio.

The SBP ranged from 133 to 371 L kg^{-1} COD with a methane content ranging from 50 to 57%. Toward the end of the experimentation, SBP rates averaged 300 L kg^{-1} COD with a methane content of 52%. The SPS was 111 g TS kg^{-1} COD. Average sludge parameters are shown in Table 1. Monitored inhibition was 26.6% in respect to measured values during the BMP test (Figure 3).

Procedure 3: precipitation of sulfides by addition of iron cations (R3)

The inhibitory effect of forming sulfides was suppressed by adding $FeCl_2$ (under the commercial product Kemira BDP 840). Daily feeding of the reactor consisted of waste biomass, precipitant BDP 840 and a 10% solution of NaOH at a ratio of 10:2.5:1.35. Daily amount of Fe^{2+} added to the bioreactor was slightly higher than the necessary stoichiometric amount for the entire precipitation of forming sulfides. The solution of NaOH acted as a pH stabilizer of the system, neutralizing the acidity of waste biomass and precipitant BDP 840.

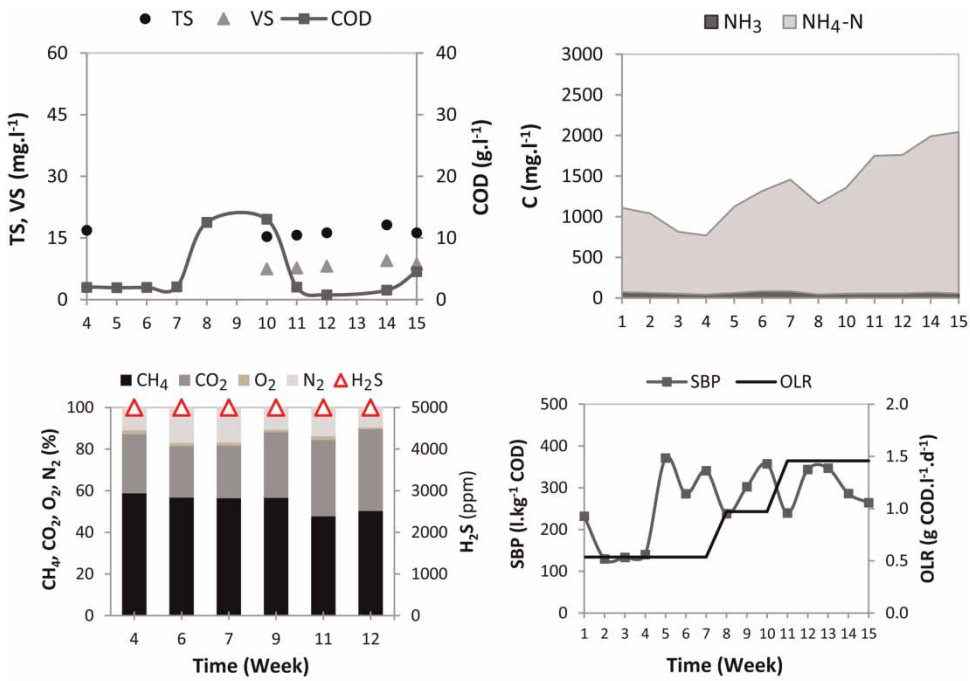


Figure 3 | Dilution with water (R2). Variation of selected parameters in bioreactor.

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The inhibitory effect of sulfides in reactor sludge was suppressed through their precipitation to ferrous salts, avoiding the formation of hydrogen sulfide in reactor sludge and its absorption into biogas. With the dosage of precipitant BDP 840, the pH was maintained neutral and the concentration of free ammonia was within a safe range (below 100 mg L^{-1}), although NH_4-N concentration was higher than 2,000 mg L^{-1} during the operation of the reactor. Fluctuations in the concentrations of COD and VFA in sludge during experimentation were caused by the increment of OLR and the amount of precipitant. However, anaerobic reactor sludge showed an ability to adapt to those changes. The concentration of PO_4-P was also maintained at lower levels than for the other studied procedures, due to the precipitation of phosphates (Figure 4).

The biogas production averaged a yield rate of 320 L kg^{-1} COD, with a 46% methane content, and very low concentrations of hydrogen sulfide (less than 200 ppm_v) toward the last weeks of the experimentation, allowing its direct combustion in cogeneration units. However, the dosage of the precipitating agent and caustic solution resulted in the gradually increment of DIS and TS of the sludge. The TS parameter was two times higher than for micro-aeration and dilution procedures. This was partially caused by dissolved and suspended inorganic ferrous salts (e.g. FeS and $Fe_3(PO_4)_2$). DIS gradually increased by 15 g L^{-1} during reactor operation up to 23.5 g L^{-1}, at the end of the experimentation. Such high levels of DIS can cause inhibition of the anaerobic process under certain conditions (Baere *et al.* 1984). Another disadvantage is the incurred costs associated with dispensing reagents and further separation and disposal of precipitates and sludge. The SPS was 400 g TS kg^{-1} COD at the end of the experimentation.

A brief summarization of selected average parameters towards the end of the experimentation can be seen in Table 2.

CONCLUSIONS

The anaerobic digestion of waste biomass in suspended-growth bioreactors indicated severe inhibition of the anaerobic processes (up to 77% after four doses of sample 1), caused by the accumulation of sulfides in reactor sludge. The procedures implemented to inhibit anaerobic reactor sludge (micro-aeration, dilution of substrate and precipitation of sulfides) significantly contributed to counter-attack the inhibitory effects caused by sulfides, involving particular advantages and disadvantages that must be considered and optimized when implementing a full-scale project. Micro-aeration was effective at reducing the concentration of sulfides in reactor sludge and hydrogen sulfide in biogas; however, this procedure may also reduce the biogas

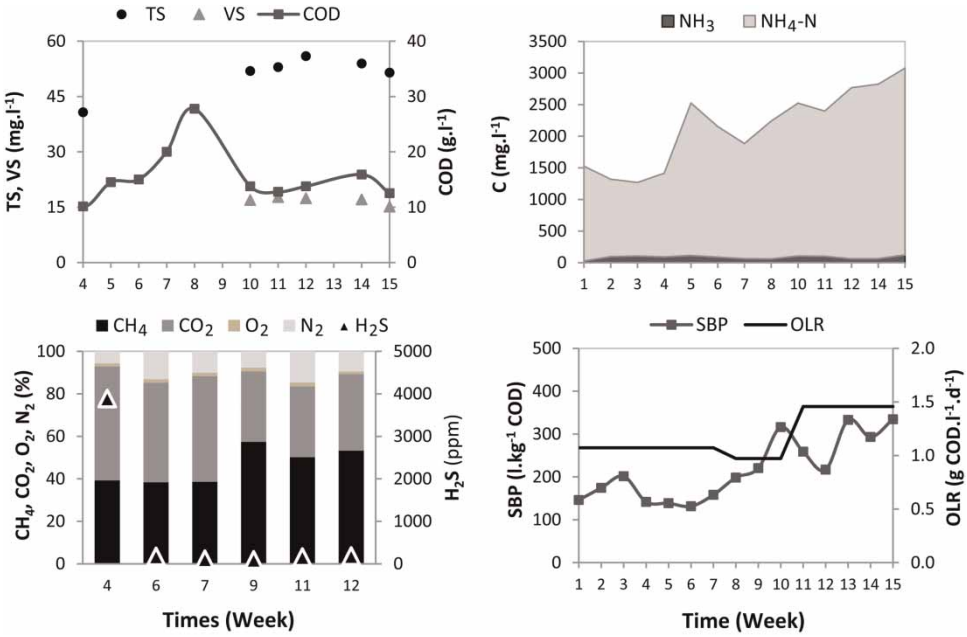


Figure 4 | Precipitation of sulfides (R3). Variation of selected parameters in bioreactor.

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Table 2 | Summarization of selected average parameters towards the end of the study

Parameter	Bioreactor	R1	R2	R3
Biogas production rate	mL mL^{-1}	29	41	44
	l kg^{-1} COD	217	300	320
Methane content	%	21	52	54
Reactor sludge (filtrate)	mg L^{-1}			
COD		6,700	3,220	13,400
VFA		5,080	2,340	7,950
NH_4-N		2,260	1,980	2,950
PO_4-P		285	270	200
DIS		7,460	4,470	23,500
Solids content of sludge	g TS L^{-1}	20.1	16.6	53.6
Specific production of sludge	kg TS kg^{-1} COD	0.133	0.111	0.400
Inhibition test according to biogas potential obtained in kinetic tests	%	48.5	26,6	25.4

production rate and the methane content in biogas, due to dilution of biogas occasioned by molecular nitrogen in air. Other operational problems can occur with the accumulation of sulfur on the surface of the anaerobic reactor, pipelines and other reactor equipment. Dilution of substrates reduced inhibition by sulfides, but the concentration of hydrogen sulfide remained very high (estimated at 18,000 ppm_v). This may cause inhibition of the process at long-term. Moreover, with this procedure, a greater amount of digestate would be produced. Precipitation of sulfides with ferrous cations was probably the most adequate procedure in terms of biogas quality and quantity. Furthermore, the operation of the anaerobic reactor during the experimentation was more stable. Nevertheless, the concentration of DIS increased throughout the operation of the reactor. This may cause inhibition of the process at long-term. Another disadvantage is the specific production of sludge, which was higher than for the other studied procedures. A combination of studied procedures and their optimization may yield better results. Despite complications caused by inhibition, waste biomass from the production of L-cystine has a relatively high potential for biogas production.

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